

Resource Reviews

Spatial Hearing: The Psychophysics of Human Sound Localization, Revised Edition Jens Blauert. Cambridge, MA: MIT Press (1997). 494 pp., ISBN 0-262-02413-6, \$38.64. Orders: (617) 253-5643.

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In 1974, Dr. Blauert first published *Spatial Hearing*. Unfortunately for those of us who do not read German, that edition was published only in German. In 1983 a new edition was published, this time with an English translation of the original three chapters from the 1974 book and a new chapter, Chapter 4, "Progress and Trends since 1972." Now, in 1997, a third edition of *Spatial Hearing* has been published with a new, fifth chapter, "Progress and Trends since 1982." That is, the latest edition contains the original three chapters from the 1974 edition, Chapter 4 from the 1983 edition, and a new chapter.

A review of this book probably should provide an opinion on two questions, "Is the material from 1974 and 1983 still valuable?" and "How valuable is the new chapter on "Progress and Trends since 1982?" To the first question, I feel the answer is a resounding yes. Anyone interested in spatial hearing should read and be familiar with the coverage provided in Chapters 1 to 4. When *Spatial Hearing* was first published, I heard many in our field who could read some German refer to the book as the "bible" on spatial hearing. When the 1983 edition was published and I could read it for the first time, I could see why the book had gained that reputation. Until *Spatial Hearing*, reviews and books on localization tended not to cover the literature worldwide. Jens Blauert's coverage of this literature on spatial hearing is complete and cannot be found anywhere else that I know of.

Although some of the research covered in the first four chapters is dated and not as useful as it might once have been, most of the material is still extremely valuable. Chapter 1 is an introduction to Dr. Blauert's view of the study of spatial hearing with a succinct introduction to the stimuli, instruments, and methods used in its study. Chapter 2 covers localization in all three spatial planes and what Dr. Blauert calls localization "blur" (in some of the literature, blur is the measure of the minimal audible angle for locating sound sources). Chapter 2 has excellent sections on the transfer functions from the outer to the middle ear, including measures of both monaural and binaural aspects of these important functions. This chapter covers localization in the median plane, including the mid-sagittal plane, and the literature up to 1972 on headphone-presented stimuli as a means to study localization (sometimes called studies of lateralization). Chapter 2 ends with a discussion of other variables that influence localization such as head motion and contributions from other neural systems, such as the vestibular system.

Chapter 3 concentrates on multiple sound sources and aspects of locating sounds in enclosed spaces (room acoustics). The concept of source coherence is introduced and used to discuss several aspects of localization when more than one sound source is present. The Law of the First Wavefront (the Precedence Effect) is described in some detail in Chapter 3. Since binaural masking involves two sounds, the signal to be detected and the masker, the literature on the binaural masking-level difference (BMLD) is reviewed. This chapter ends with a consideration of localization in diffuse sound fields. Chapter 4 updates many of the topics of Chapters 2 and 3 with data and theories developed between 1972 and 1982. New topics such as models of binaural processing and an interesting section on applications also are covered in Chapter 4.

The new chapter (Chapter 5) in the latest edition of *Spatial Hearing* covers two main issues: Head Related Transfer Functions (HRTFs) and their use in virtual environments, and new research about the precedence effect. Anyone with any interest in echo processing and the precedence effect must read this section of the new chapter. The new chapter provides an excellent overview of the work on and the promise for auditory virtual environments. In combination with the material covered in Chapter 2 and updated in Chapter 4, the book provides a strong review of the fundamentals of the HRTF and their use in understanding spatial hearing. However, the coverage of recent advances in the study of auditory virtual worlds is more general and does not provide the in-depth coverage that most other topics receive.

Chapters 1 through 4 have been valuable to auditory scientists for many years and will continue to be so for several more years. If there is any disappointment in the new edition, it is that Dr. Blauert did not provide the same level of review and explanation for the literature since 1982 as he did for that before 1982. He clearly has provided that type of review with respect to the latest work on the precedence effect. However, the latest edition does not provide a thorough review of recent work on HRTFs, new developments in binaural models, and concepts such as binaural interference (an expansion of the Blauert's earlier concept of coherence). I also have a small quibble about concluding that the new evidence that precedence is affected by past stimulus conditions necessitates a cognitive explanation of precedence. I believe that at the present time more data and theories of precedence are needed, and both cognitive and noncognitive explanations need to be explored.

In conclusion, if you are a student or a professional in the field of hearing and you do not already have a copy of *Spatial Hearing*, you need to purchase it. If you already have a copy and you are interested in spatial hearing, especially in the precedence effect, I highly recommend that you purchase the new edition. Otherwise, you might look at a copy before you decide whether you want to replace your older edition.